Animal Pose Estimation

using Physics-Informed Neural Networks

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Agenda

Why Animal Pose Estimation?

Introduction

Background

Pose estimation has thrived for humans with large datasets and deep CNNs — but animals remain a frontier due to 850+ distinct species, varied body plans, and sparse annotated video data.

Problem

A single neural network must generalize across species captured in the wild under occlusion, scale changes, and lighting variation and handle temporal video sequences, not just still images.

Our Approach

Physics-Informed Neural Network (PINN) with bone-length rigidity, temporal smoothness, and joint-angle plausibility constraints trained on the Animal Kingdom benchmark.

Objectives

PINN Architecture

Design a multi-species, temporally-aware pose estimation model with physics-informed constraints.

Physics Loss Terms

Formulate and validate bone-length rigidity, temporal smoothness, and joint-angle plausibility losses.

Benchmark Target: PCK > 70%

Train on Animal Kingdom (850+ species, 23 keypoints) and exceed 70% PCK on the validation set.

YOLOv8 Integration

Integrate YOLOv8 as a detection frontend to improve inference on small-animal frames.

Methodology — Dataset & Architecture

Animal Kingdom Dataset

850+ Animal Species	**23** Keypoints / Frame
7 Taxonomic Classes	**7,157** Video Clips
83,187 Total Annotations	**69,121** Valid Triplets

PoseCNN Architecture

Input

Temporal triplet (t0-> t1-> t2) — 256×256×3

Backbone

ResNet-18 (ImageNet pretrained) — 13.9M params

Decoder

4× Transposed Conv layers — upsampled feature maps

Output

Heatmaps (B, 23, H, W) → Soft-argmax → 2D coords

Physics-Informed Loss Function

```
L_total = L_data + 0.5·L_bone + 0.1·L_smooth + 0.05·L_angle
```

L_data
λ = 1.0

Data Loss

Visibility-masked MSE between predicted and ground-truth 2D keypoint coordinates. Invisible keypoints (visibility=0) excluded.

L_bone
λ = 0.5

Bone Rigidity

MSE between predicted bone lengths and reference lengths from training set. Enforces anatomical rigidity across frames.

L_smooth
λ = 0.1

Temporal Smoothness

L2 penalty on temporal difference of keypoint positions between consecutive triplet frames. Penalizes sudden jumps.

L_angle
λ = 0.05

Joint realism

Penalizes joint angles outside anatomically plausible ranges estimated from training data.

Exploration & Challenges

Failed Attempt 1: Wrong Dataset

COCO-format wildlife dataset (648 images, 17 keypoints) — no temporal structure. Incompatible with PINN triplet loss. Dataset scrapped entirely.

Failed Attempt 2: TRI-MOUSE

Validated triplet construction on DLC-format mouse dataset. Too narrow in scope — single species, lab setting. Adapted to Animal Kingdom.

Kaggle Session Limits — Solved

40-min idle timeout + 12-hr limit. Fixed with num_workers=4, pin_memory=True, 5K triplets/epoch (~70s/epoch). Added resume checkpointing.

Training Success ✓

50 epochs completed successfully. PCK rose from ~18% at epoch 1 to 85.4% at epoch 49. All loss components converged cleanly.

Results & Analysis

85.4%

Best PCK
(Epoch 49/50)

0.0041

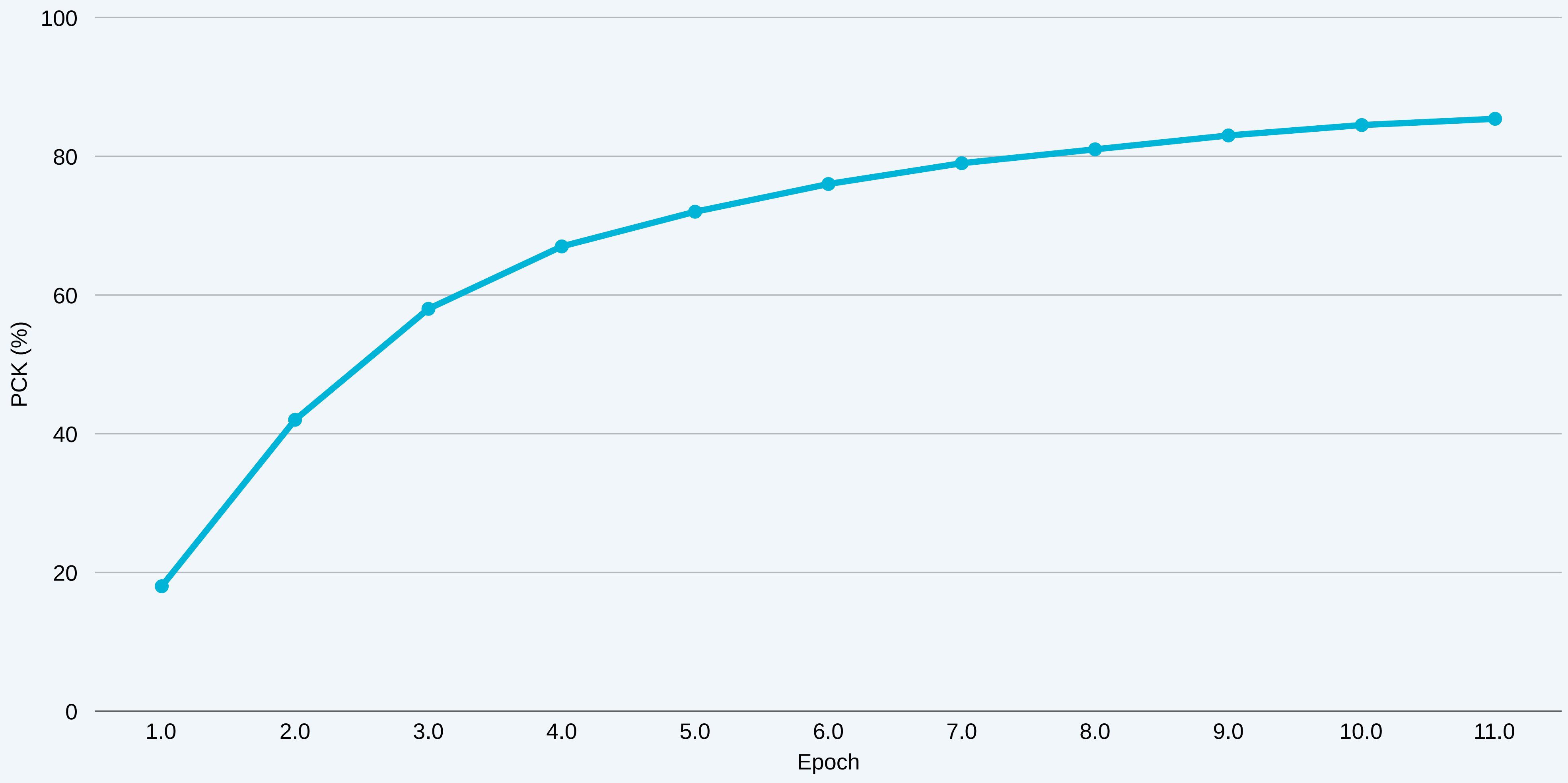
Final Val Loss
(from 0.027)

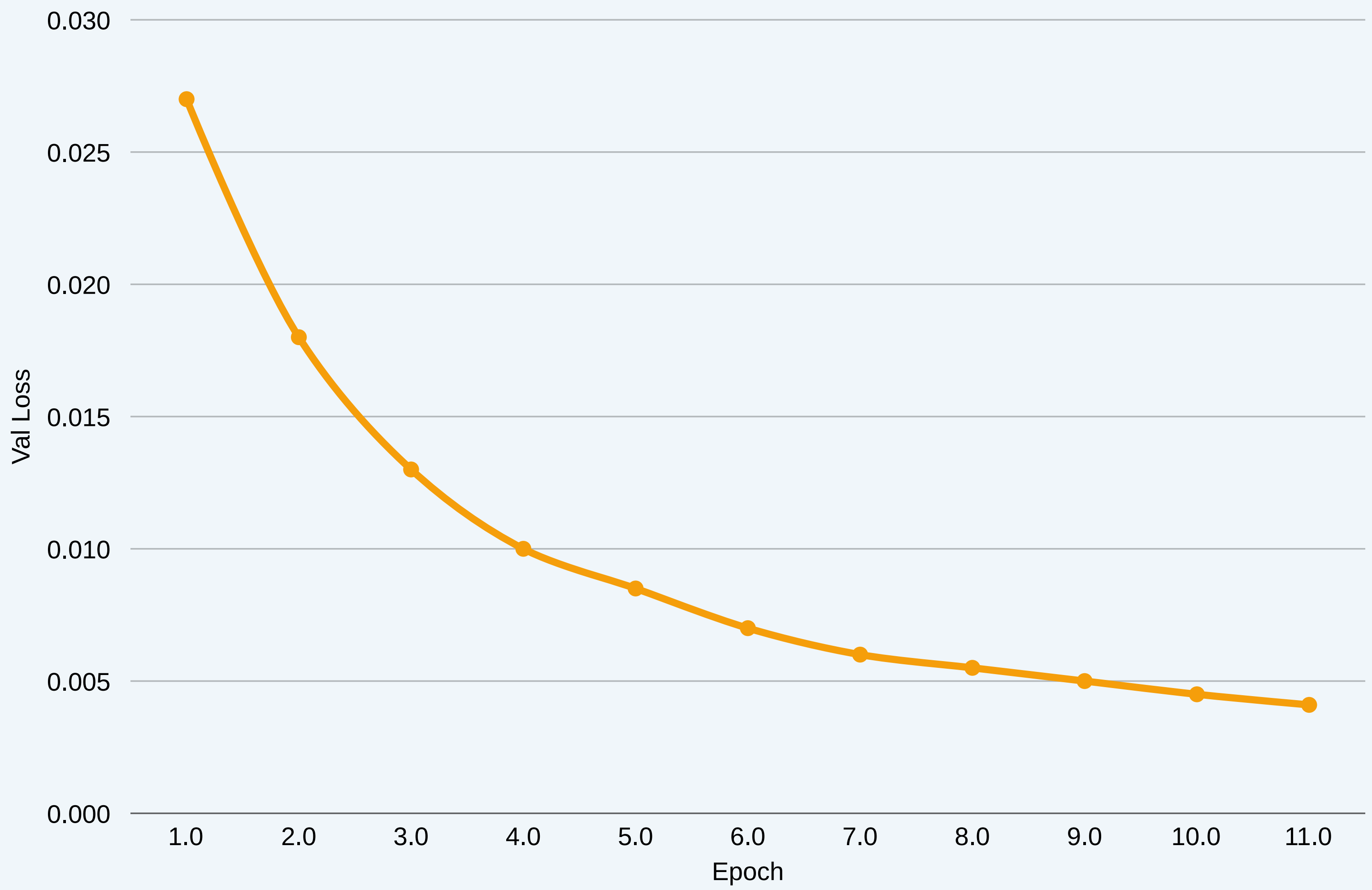
850+

Species
Covered

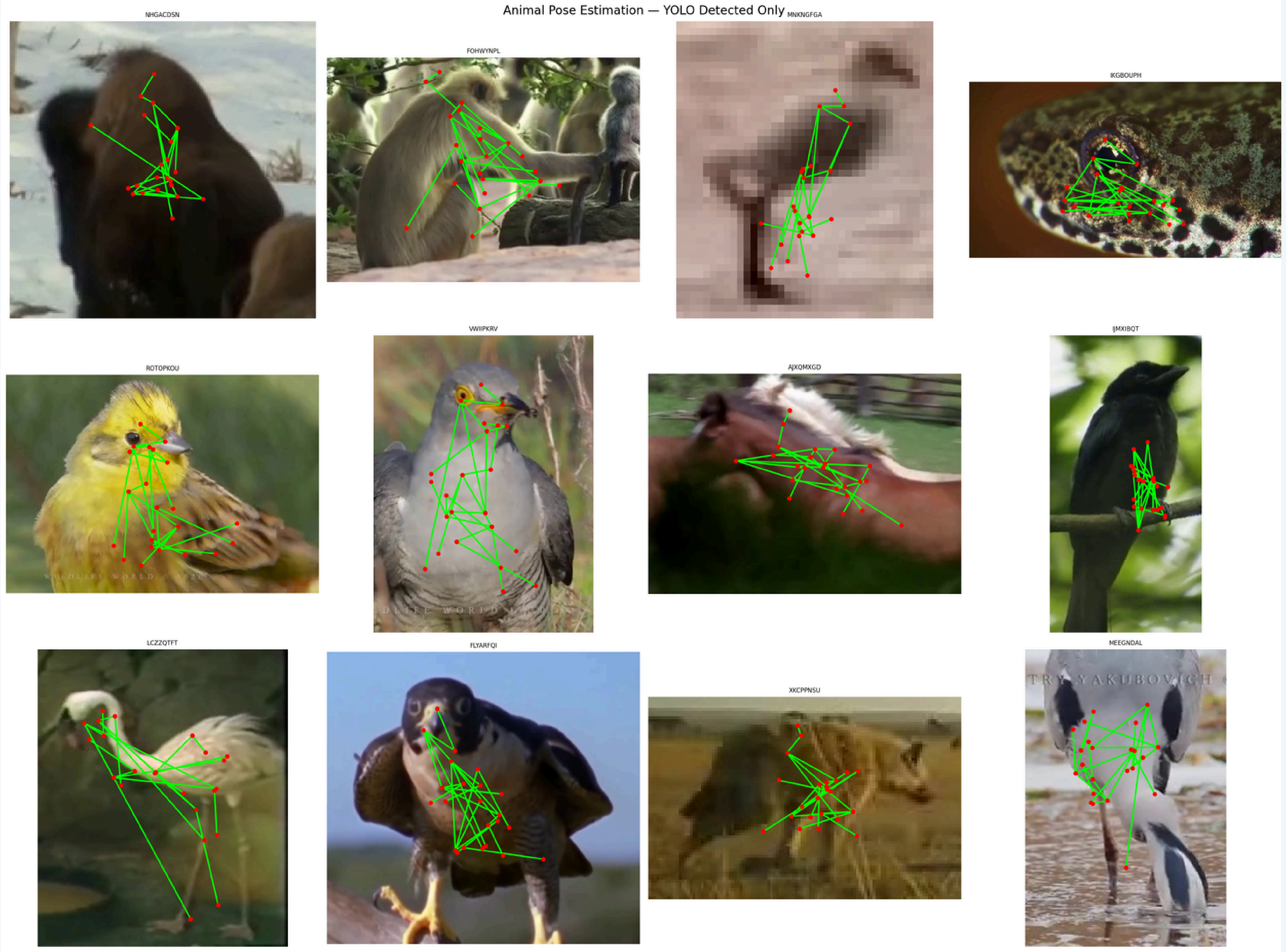
69,121

Valid
Triplets





Qualitative Results – Skeleton Visualization



Qualitative Results & YOLOv8 Integration

Performance by Group

Birds	Strong
Mammals	Good
Insects	Weaker

YOLOv8 → PoseCNN Pipeline

1 YOLOv8n detects largest animal in frame

2 Bounding box returned with 10% padding

3 Crop resized to 256×256 and fed to PoseCNN

4 Keypoints mapped back to original image space

Result: Keypoints spread anatomically vs. clustered at center without YOLO.

Discussion

✓ What Worked

- PINN loss enforced temporal consistency — smooth, plausible poses across all 3 frames.
- Warm-up strategy (head-only, 5 epochs) stabilized early training effectively.
- YOLOv8 + PoseCNN pipeline improved visual results with zero retraining cost.

⚠ Limitations

- No cross-dataset or held-out species evaluation — same distribution tested only.
- 5K triplets/epoch subsampling — model has seen only a fraction of full dataset.
- YOLOv8 not trained on all 850 species — exotic / small animals degrade results.

Conclusion & Future Work

Key Takeaway

PINN constraints are a viable, lightweight addition to animal pose estimation — achieving 85.4% PCK across 850 species.

Future Work

- 💻 Train on full 69K triplets with multi-GPU DataParallel
- 🧠 Replace ResNet-18 with HRNet or ViT-Pose backbone
- 🐦 Fine-tune YOLOv8 on Animal Kingdom species
- 🧬 Species-conditioned pose estimation via species embeddings
- 🌍 Zero-shot generalization on held-out unseen species
- 📐 3D pose lifting from 2D via depth estimation / multi-view

Thank You

Key Resources

Training Notebook
kaggle.com/code/bhavika2008/notebook8917a292bb

Image Dataset
kaggle.com/datasets/bhavika2008/ak-images

Code & Annotations
kaggle.com/datasets/bhavika2008/ak-code-ann

Animal Kingdom
github.com/sutdcv/Animal-Kingdom